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A test battery for evaluating hop performance in patients with an ACL injury and patients who have undergone ACL reconstruction

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Abstract The purpose of this study was to develop a test battery of hop tests with high ability to discriminate (i.e. high test–retest reliability, sensitivity, specificity and accuracy) between the hop performance of the injured and the uninjured side in patients with an ACL injury and in patients who have undergone ACL reconstruction. Five hop tests were analysed: three maximum single hop tests and two hop tests while developing fatigue. Fifteen healthy subjects performed the five hop tests on three separate occasions in a test–retest design. Thirty patients, mean 11 months after an ACL injury and 35 patients, mean 6 months after ACL reconstruction were tested. ICC values ranged from 0.85 to 0.97 for the five hop tests, indicating that all the tests had high test–retest reliability. Sixty-seven percent to 100% of the healthy subjects had normal symmetry (i.e. <10% side-to-side difference) in the five hop tests. Abnormal symmetry in the five hop tests ranged from 43 to 77% for patients with an ACL injury and from 51 to 86% for patients who had undergone ACL reconstruction respectively. The three tests with the highest ability to discriminate hop performance were chosen for the test battery; they were the vertical jump,

the hop for distance and the side hop. The test battery revealed a high level of sensitivity and accuracy in patients with an ACL injury (87 and 84%) and in patients who had undergone ACL reconstruction (91 and 88%), when at least one of the three tests was classified as abnormal. To summarise, the test battery consisting of both maximum single hop performances: the vertical jump and the hop for distance and hop performance while developing fatigue: the side hop, produced high test–retest reliability, sensitivity and accuracy. Further, the test battery produced higher values compared with any of the three hop tests individually revealing that only one out of ten patients had restored hop performance 11 months after an ACL injury and 6 months after ACL reconstruction. It is concluded that this test battery showed a high ability to discriminate between the hop performance of the injured and the uninjured side both in patients with an ACL injury and in patients who have undergone ACL reconstruction.

Keywords Anterior cruciate ligament · Knee · Rehabilitation · Hop test

Introduction

One difficult challenge in the rehabilitation of anterior cruciate ligament (ACL) injury and reconstruction is to determine when it is safe to return to strenuous physical activities. Single-leg hop tests are commonly used to evaluate functional performance after an ACL injury [1–7] or reconstruction [8–19]. To some degree, hop tests have been able to discriminate between the hop performance of the injured and uninjured side in patients with an ACL injury and between controls and patients [1, 2, 4, 7, 10]. There is also some evidence that single-leg hop tests are important when trying to predict whether patients will have future difficulty in terms of recurrent knee instability after an ACL injury [3].

The ratio between the involved and uninjured leg has been the most frequently reported criterion for determining normal or abnormal hop test scores [1–6, 8–10, 13, 15, 17–21]. It has been suggested that the normal ratio in healthy subjects is greater than or equal to 85% [1, 21] or 90% [10, 20].

Several hop tests are described in the literature, including various single-leg hop tests for distance, time [1–6, 8–11, 13–21] and height [1, 13, 15, 20, 21]. The evaluation of functional outcome after an ACL injury is often limited to only one test, the single-leg hop for distance [9, 14, 17, 18]. However, the reported sensitivity for detecting functional limitations associated with ACL deficiency with the single-leg hop for distance test is relatively low, ranging from 38 to 52% [4, 5, 7]. Noyes et al. [5] combined two horizontal hop tests, a single-leg hop for distance and a timed hop in order to increase the sensitivity. Itoh et al. [4] argued that hop tests should involve more twisting and cutting movements and evaluated the figure-of-eight hop test, the up-down hop test and the side hop test. When these three tests and the single-leg hop for distance test were combined in a test battery, a sensitivity of 82% was obtained. Sports injuries often tend to occur at the end of a sporting event, when a participant is fatigued [22–24]. None of the tests recommended by either Noyes et al. [5] or Itoh et al. [4] is, however, performed under fatigued conditions. Augustsson et al. [8] reported an improved sensitivity level when testing hop performance under fatigued conditions. All patients in the study by Augustsson et al. [8] were classified, 11 months after ACL reconstruction, as having normal hop capacity when performing a single leg hop for distance non-fatigued. After a fatiguing quadriceps muscle exercise only one-third of the patients were classified as having normal hop performance.

The purpose of this study was to develop a test battery of hop tests with a high ability to discriminate (i.e. high test–retest reliability, sensitivity, specificity and accuracy) between the hop performance of the injured and the

uninjured side in patients with an ACL injury and in patients who have undergone ACL reconstruction.

We hypothesise that a test battery evaluating different hop qualities, i.e. maximum single hop performance, as well as hop performance while developing fatigue, will increase the opportunity to detect discrepancies in hop performance (i.e. increase the test sensitivity) compared with using only a single hop test.

Materials and methods

Subjects

Three groups of subjects participated in this study, healthy individuals, patients with an ACL injury and patients who had undergone ACL reconstruction.

The patients physical activity levels prior to their ACL injury and at the test occasion were documented using the Tegner score [25]. The Tegner score is an activity grading scale, where activities of daily living, recreation, competitive sports and work are graded numerically from 1 to 10. One represents the least strenuous knee activity and ten is hard strenuous knee activity such as rugby and international soccer. The score was modified in the year 2000 but this version has not been published. The modified version was used in the present study with the permission of the authors. To complement the Tegner score with the subjects' intensity and frequency in participation in physical activity a four grade *physical activity scale* was constructed, using a validated score for elderly people [26] as a model. After discussions in an expert group consisting of experienced physical therapists and orthopaedic surgeons good face validity of the new *physical activity scale* was assured. On the physical activity scale the subjects made their own judgment on how vigorously and frequent they participated in physical activity at the *present time* as well as *prior* to their knee injury. The four grades in this physical activity scale were:

1. Hardly any physical activity at all.
2. Light physical activity a few hours a week.
3. Somewhat strenuous physical activity 2–3 h a week.
4. Hard strenuous physical activity on a regular basis.

A convenience sample of nine male and six female healthy subjects were recruited. The subjects had no history of back, hip, knee or ankle dysfunction and were tested on three separate occasions in a test–retest design.

Thirty patients, 18 males and 12 females, with an ACL injury were tested. An interval of 4–44 months elapsed between the index knee injury and the test occasion. The inclusion criteria consisted of a positive anterior drawer and Lachman's test, performed by

experienced orthopaedic surgeons and verified by the patient's physiotherapist, history of knee injury and subjective giving way, no acute pain or swelling and no prior surgical procedure in either leg.

Thirty-five patients, 25 males and 10 females, who had undergone ACL reconstruction, were tested. The patients were included in this study if they met the following criteria, no acute pain or swelling and no prior surgical procedure in either leg. The operated patients were tested 6 months after ACL reconstruction. Descriptive data for all subjects are presented in Table 1.

All patients in this study were recruited from the same physiotherapy clinic and underwent supervised physiotherapy according to a criterion based protocol. The patients trained two to three sessions per week (initially combined with a home based program). Wet-vest training and functional group training were offered as additional sessions. Full immediate weight bearing was allowed post operatively. Early range of motion exercises was encouraged. Both closed and open kinetic chain exercises were used to restore muscle strength of lower extremity muscles. Also neuromuscular training (balance, proprioception and plyometric exercises) were incorporated with gradually increased load and complexity.

Prior to any testing, written informed consent forms were signed by all the participants. Approval for the study was obtained from the Human Ethics Committee at Göteborg University, Sweden.

Procedure

The healthy subjects performed five single-leg hop tests in both legs at three different test sessions in a test-retest

design. An interval of 3–13 days elapsed between test occasion 1 and 2 and an interval of 3–19 days between test occasion 2 and 3. Subjects were asked not to participate in strenuous physical activities the day before testing. For the healthy subjects, the order of the tests and the leg that was first tested were randomised, whereas the test order was pre-determined for the patients. The patients performed the tests in the same order as they are presented. The practice and test trials were performed using the uninjured leg first, followed by the injured leg. The patients were thoroughly familiarised at the physiotherapy clinic prior to the test session. All the tests were supervised by the same test leader. Verbal encouragement was used and athletic footwear were standardised.

Before testing, the subjects completed a warm-up consisting of 5 min of stationary cycling, two times 10 squats, two times 10 toe rises and warm-up jumps.

The following single-leg hop tests were used: (1) vertical jump, (2) hop for distance, (3) drop jump followed by a double hop for distance, (4) square hop and (5) side hop. The tests were chosen on the basis of hop tests commonly described in the literature [1–6, 8–21], as well as clinical experience, and were designed to reflect various hop qualities. The tests should also be easy to administer in a clinical setting. The subjects performed three to five practice trials followed by three maximum approved trials for the vertical jump, the hop for distance and the drop jump, followed by a double hop for distance. However, if the subjects increased their hop performance in all three hops, additional hops were performed until no increase was seen. The square hop and the side hop were tested once. Three minutes of rest were used between each hop test. The best trial for each leg in each test was used for data analysis. The hop tests were videotaped for subsequent control of test procedures.

Table 1 Descriptive characteristics of healthy subjects and patients

	Age (years)	Height (cm)	Weight (kg)	Tegner before injury	PAS before injury	Tegner at test occasion	PAS at test occasion	Weeks injury/op-test
Healthy subjects								
Male (<i>n</i> = 9)	29 ± 5	181 ± 6	84 ± 10					
Female (<i>n</i> = 6)	26 ± 4	168 ± 8	61 ± 6					
All subjects (<i>n</i> = 15)	28 ± 4	175 ± 9	75 ± 14					
ACL injury								
Male (<i>n</i> = 18)	28 ± 7	181 ± 8	91 ± 17	7.5 (2.0)	4.0 (1.0)	4.0 (2.2)	2.5 (1.0)	50 ± 48
Female (<i>n</i> = 12)	36 ± 8	167 ± 5	71 ± 12	4.5 (3.5)	3.0 (1.0)	3.0 (1.0)	3.0 (1.0)	45 ± 35
All subjects (<i>n</i> = 30)	31 ± 9	175 ± 9	83 ± 18	7.0 (3.0)	3.0 (1.0)	4.0 (1.2)	3.0 (1.0)	48 ± 43
ACL reconstruction								
Male (<i>n</i> = 25)	27 ± 8	180 ± 8	81 ± 11	8.0 (2.0)	3.0 (1.0)	4.0 (3.5)	3.0 (1.5)	28 ± 2
Female (<i>n</i> = 10)	27 ± 7	167 ± 5	60 ± 5	7.0 (2.5)	3.5 (1.0)	3.0 (1.0)	3.0 (1.0)	28 ± 2
All subjects (<i>n</i> = 35)	27 ± 7	177 ± 10	75 ± 13	8.0 (2.0)	3.0 (1.0)	4.0 (2.0)	3.0 (1.0)	28 ± 2

All values are expressed as means (±SD), except for Tegner activity score and Physical Activity Scale (PAS), which is expressed as median values (interquartile range)

Hop tests

Vertical jump (Fig. 1a)

The vertical jump test was performed as a counter-movement jump. The starting position was an upright position with the hands placed behind the back. The subjects quickly bent their knee as much as desired and then immediately jumped upwards, attempting to maximise the height jumped. A computerised system (MuscleLab, Ergotest Technology) using a field of infrared light (approximately 10 mm above the floor), serving as a “contact mat”, made it possible to measure the flight time. The system then converted the flight time into jump height in centimetres.

Hop for distance (Fig. 1b)

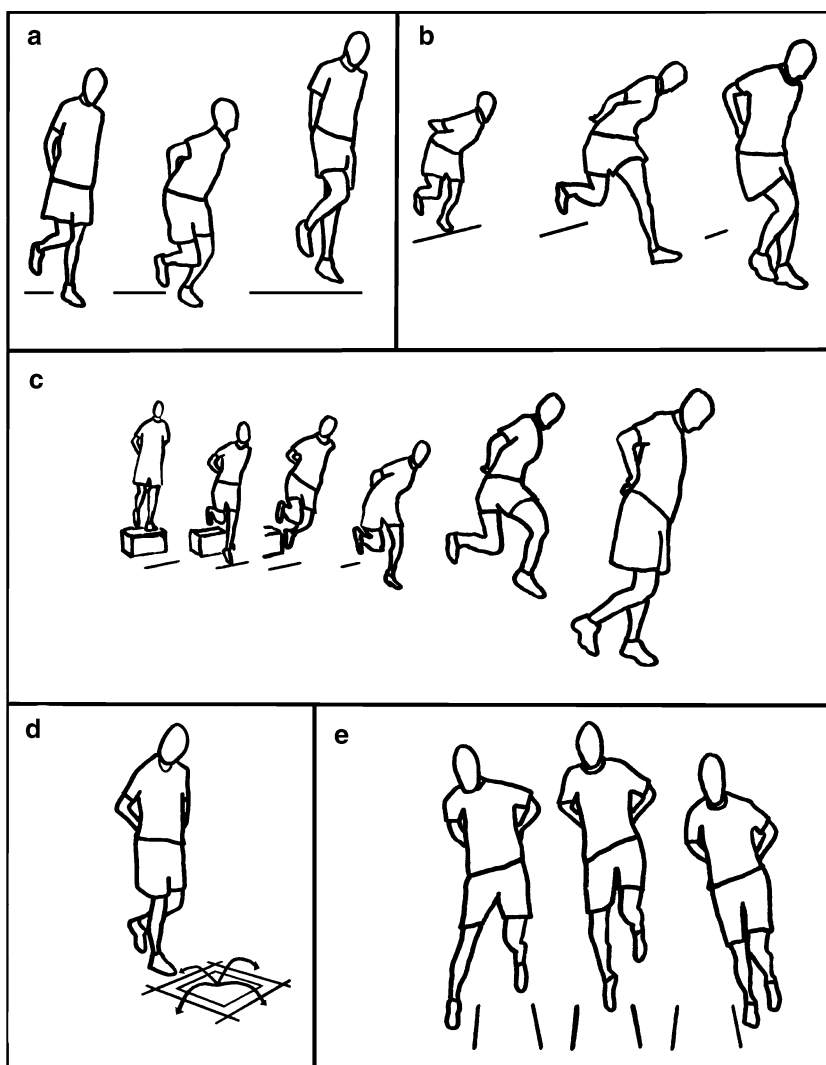
The subjects stood on the test leg and then hopped as far as possible and landed on the same leg. Free leg swing

was allowed. The hands were placed behind the back. The subjects were instructed to perform a controlled, balanced landing and to keep the landing foot in place (i.e. no extra hops were allowed) until (2–3 s) the test leader had registered the landing position. Failure to do so resulted in a disqualified hop. The distance was measured in centimetres from the toe at the push-off to the heel where the subject landed.

Drop jump followed by a double hop for distance (Fig. 1c)

The starting position was standing on the leg to be tested on a box, at a height of 30 cm, with the hands behind the back. Forty-five centimetres in front of the box, a strip of tape marked the starting line. The subjects jumped down on one leg, without crossing or touching the starting line, and then immediately performed two one-legged maximum hops forward. The subjects were instructed to

Fig. 1 Five single leg hop tests were used: **a** vertical jump, **b** hop for distance, **c** drop jump followed by a double hop for distance, **d** square hop and **e** side hop



perform a controlled, balanced landing and to keep the landing foot in place (i.e. no extra hops were allowed) until (2–3 s) the test leader had registered the landing position. Failure to do so resulted in a disqualified hop. The distance was measured in centimetres from the starting line to the heel where the subject landed.

Square hop (Fig. 1d)

The subjects stood on the leg to be tested, with their hands behind their back, outside a 40×40 cm square marked with tape on the floor. A 10 cm frame was also marked around the square with tape. For the right leg, the subjects were instructed to jump clockwise in and out of the square as many times as possible during a period of 30 s. The number of successful jumps performed, without touching the taped frame, was recorded. Touching the taped frame was recorded as an error and, if more than 25% of the jumps had errors, a second trial of 30 s was performed after a 3-min rest period. For the left leg, the subject performed the test in a counter-clockwise mode. This test was modified from Östenberg et al. [21].

Side hop (Fig. 1e)

For the side hop test, the subjects stood on the test leg, with their hands behind their back, and jumped from side to side between two parallel strips of tape, placed 40 cm apart on the floor. The subjects were instructed to jump as many times as possible during a period of 30 s. The number of successful jumps performed, without touching the tape, was recorded. Touching the tape was recorded as an error and, if more than 25% of the jumps had errors, a second trial of 30 s was performed after a 3-min rest period. This test was modified from Itoh et al. [4].

Statistical analysis

Test–retest analysis was determined by an intraclass correlation coefficient (ICC) with a 95% confidence interval (CI) [27]. The methodological error in percent was calculated according to the formula [28]:

$$[e/m] \times 100 = \% \text{ difference between test}$$

e	the square root of $[\sum (d^2)/2n]$
d	difference between tests
n	number of subjects
m	mean value.

The mean and standard deviation were calculated for the jump tests.

Wilcoxon's signed rank test was used for dependent values and the Mann–Whitney U test was used for

independent values. The Spearman Rank was used to study the relationship between dependent variables. The level of significance was set at $P \leq 0.05$.

The lower limb symmetry index (LSI) was calculated to determine whether a side-to-side leg difference was classified as normal or abnormal. The LSI is defined as the ratio of the involved limb score and the uninvolved limb score expressed in per cent (involved/uninvolved $\times 100 = \text{LSI}$). In this study, an LSI greater than or equal to 90% [10, 20] was classified as normal.

Sensitivity, specificity and accuracy were calculated for the five hop tests and for the test battery using the following definitions.

Sensitivity (= number of patients classified as abnormal/total number of patients) expresses the percentage probability that the tests would demonstrate an abnormal LSI in the patients [29].

Specificity (= number of healthy subjects classified as normal/total number of healthy subjects) expresses the percentage probability that the tests would demonstrate a normal LSI in the normal subjects [29].

Accuracy (= number of patients classified as abnormal + number of healthy subjects classified as normal/total number of patients and healthy subjects) is defined as the percentage probability that the tests would demonstrate a normal LSI in the normal subjects and an abnormal LSI in the patients [29].

A factor analysis with a principal axis factoring, varimax rotation, and for two factors was applied for evaluation of the structure of the test battery.

Results

Test–retest reliability for the five hop tests

Significant differences were found between test occasion 1 and 2 for male subjects and for male and female subjects combined in the hop for distance ($P=0.02$) and the square hop ($P=0.001$). A significant difference was also found between test occasion 1 and 2 for all subjects in the side hop ($P=0.01$). No significant differences were found between test occasion 2 and 3 in any of the five hop tests, as shown in Table 2.

The ICC, 95% CI and methodological error in percent and in absolute values are presented in Table 2. In the five tests, the ICC between test occasion 2 and 3 ranged from 0.85 to 0.97 and the methodological error ranged from 3 to 6%.

Comparison of hop performance between healthy male and female subjects

A comparison between healthy male and female subjects' hop performance in the five hop tests on test

Table 2 Test-retest reliability between test occasion 1 and 2 (Test₁₋₂) and test occasion 2 and 3 (Test₂₋₃)

Test		<i>P</i> -value	ICC	95% CI	Methodological error	
					%	cm or jumps
Vertical jump (cm)						
Male (<i>n</i> = 9)	Test ₁₋₂	0.08	0.83	0.61–0.93	8	1.4
	Test ₂₋₃	0.85	0.95	0.88–0.98	4	0.7
Female (<i>n</i> = 6)	Test ₁₋₂	0.72	0.95	0.84–0.99	5	0.7
	Test ₂₋₃	0.82	0.97	0.90–0.99	4	0.6
All subjects (<i>n</i> = 15)	Test ₁₋₂	0.11	0.89	0.79–0.95	7	1.2
	Test ₁₋₂	0.94	0.97	0.93–0.98	4	0.6
Hop for distance (cm)						
Male (<i>n</i> = 9)	Test ₁₋₂	0.02	0.91	0.78–0.97	3	4.2
	Test ₂₋₃	0.20	0.86	0.67–0.95	3	4.6
Female (<i>n</i> = 6)	Test ₁₋₂	0.35	0.88	0.64–0.96	3	4.7
	Test ₂₋₃	0.08	0.98	0.93–0.99	2	2.1
All subjects (<i>n</i> = 15)	Test ₁₋₂	0.02	0.94	0.88–0.97	3	4.3
	Test ₂₋₃	0.07	0.95	0.90–0.98	3	3.7
Drop jump followed by a double hop for distance (cm)						
Male (<i>n</i> = 9)	Test ₁₋₂	0.07	0.80	0.54–0.92	3	9.6
	Test ₂₋₃	0.41	0.85	0.65–0.94	3	9.6
Female (<i>n</i> = 6)	Test ₁₋₂	0.43	0.67	0.19–0.89	6	15.0
	Test ₂₋₃	0.13	0.91	0.73–0.97	3	8.3
All subjects (<i>n</i> = 15)	Test ₁₋₂	0.07	0.85	0.72–0.93	4	12.1
	Test ₂₋₃	0.13	0.93	0.85–0.96	3	8.7
Square hop (jumps)						
Male (<i>n</i> = 9)	Test ₁₋₂	0.00	0.64	0.18–0.87	13	6.9
	Test ₂₋₃	0.09	0.78	0.50–0.91	5	3.1
Female (<i>n</i> = 6)	Test ₁₋₂	0.06	0.55	0.01–0.85	12	6.9
	Test ₂₋₃	0.48	0.89	0.67–0.97	5	3.2
All subjects (<i>n</i> = 15)	Test ₁₋₂	0.00	0.58	0.25–0.78	13	6.9
	Test ₂₋₃	0.08	0.85	0.71–0.93	5	3.1
Side hop (jumps)						
Male (<i>n</i> = 9)	Test ₁₋₂	0.06	0.72	0.39–0.88	9	4.5
	Test ₂₋₃	0.61	0.78	0.51–0.91	5	3.0
Female (<i>n</i> = 6)	Test ₁₋₂	0.10	0.87	0.62–0.96	14	5.5
	Test ₂₋₃	0.58	0.95	0.83–0.98	8	3.5
All subjects (<i>n</i> = 15)	Test ₁₋₂	0.01	0.87	0.75–0.94	10	4.8
	Test ₂₋₃	0.95	0.93	0.87–0.97	6	3.2

P-values, intraclass correlation coefficients (ICC), 95% confidence intervals (CI) and the methodological errors are given in percent and in absolute values

occasion 3 is presented in Table 3. Significant differences were found between males and females in the vertical jump ($P=0.03$), the hop for distance ($P=0.03$) and the drop jump followed by a double hop for distance ($P=0.01$) but not in the square hop and the side hop. No significant difference was, however, found in any of the tests when comparing the absolute values for side-to-side difference in males and females. Further analyses were therefore conducted, regardless of gender.

Hop performance in patients

A significant difference between the injured and uninjured side was found in all the hop tests except for the square hop tests in patients with an ACL injury and in

all the tests in patients who had undergone ACL reconstruction (Table 4).

Comparison of hop performance between healthy subjects and patients

Patients with an ACL injury and patients who had undergone ACL reconstruction had a significantly larger side-to-side difference in all the tests except for the square hop compared with healthy subjects (Table 5). The results of the square hop test did not reveal a significant difference between healthy subjects and patients when comparing the side-to-side difference.

A normal LSI for the five hop tests ranged from 67 to 100% in the healthy subjects, from 23 to 57% in the patients with an ACL injury and from 14 to 49% in the

Table 3 Means, standard deviations (SD) in the five hop tests on test occasion 3 for the right and left leg

Test	All subjects (n = 15)	Male (n = 9)	Female (n = 6)	P-value
Vertical jump (cm)	16.7 ± 3.7	18.1 ± 3.3	15.0 ± 3.7	< 0.03
Hop for distance (cm)	151 ± 16	160 ± 11	137 ± 13	< 0.01
Drop jump followed by a double hop for distance (cm)	304 ± 34	320 ± 27	281 ± 30	< 0.01
Square hop (jumps)	62 ± 7	62 ± 6	60 ± 9	NS
Side hop (jumps)	50 ± 13	55 ± 6	41 ± 16	NS

P-values between males and females among healthy subjects are given (NS = not significant)

patients who had undergone ACL reconstruction (Fig. 2).

Sensitivity, specificity and accuracy of the five hop tests

Table 6 shows the sensitivity, specificity and accuracy of the five hop tests. *Sensitivity* ranged from 43 to 77% in the patients with an ACL injury. Sensitivity in the patients who had undergone ACL reconstruction ranged from 51 to 86%. *Specificity* ranged from 67 to 100% in the five hop tests. *Accuracy* ranged from 58 to 80% in the patients with an ACL injury. Accuracy in the patients who had undergone ACL reconstruction ranged from 56 to 86%.

Correlation between the five hop tests

The correlation in healthy subjects and patients ranged from $r_s = 0.70$ to $r_s = 0.94$ ($P = 0.01$) between the three maximum hop tests (vertical jump, hop for distance and drop jump followed by a double hop for distance). The correlation in healthy subjects and patients ranged from $r_s = 0.72$ to $r_s = 0.92$ ($P = 0.01$) between the two endurance hop tests (square hop and side hop). The correlation between the three maximum hop tests and the two endurance hop tests ranged from $r_s = 0.29$ to $r_s = 0.57$ ($P = 0.05$) in the healthy subjects and from $r_s = 0.62$ to $r_s = 0.88$ ($P = 0.01$) in the patients.

Table 4 Means, standard deviations (SD) in the five hop tests in patients with ACL injury and patients who have undergone ACL reconstruction (ACL recon.)

P-values between the injured and the uninjured sides are given

Test	ACL injury (n = 30)		P-value	ACL recon (n = 35)		P-value
	Injured	Uninjured		Injured	Uninjured	
Vertical jump (cm)	11.5 ± 5.4	14.5 ± 5.4	0.05	13.3 ± 5.0	17.5 ± 4.6	< 0.00
Hop for distance (cm)	115 ± 39	135 ± 29	< 0.04	128 ± 28	148 ± 23	< 0.00
Drop jump followed by a double hop for distance (cm)	225 ± 83	270 ± 55	< 0.03	253 ± 56	297 ± 48	< 0.00
Square hop (jumps)	38 ± 18	46 ± 15	NS	49 ± 17	57 ± 12	< 0.03
Side hop (jumps)	28 ± 17	39 ± 16	< 0.01	39 ± 16	49 ± 13	< 0.00

Factor analysis on the five hop tests

The factor analysis on the healthy subjects as well as the factor analysis on the patients divided the five hop tests in *factor one*: maximal hop tests (vertical jump, hop for distance and drop jump followed by a double hop for distance) and *factor two*: endurance hop tests (square hop and side hop).

The test battery

The following three tests were chosen for the test battery: (1) the vertical jump, (2) the hop for distance and (3) the side hop.

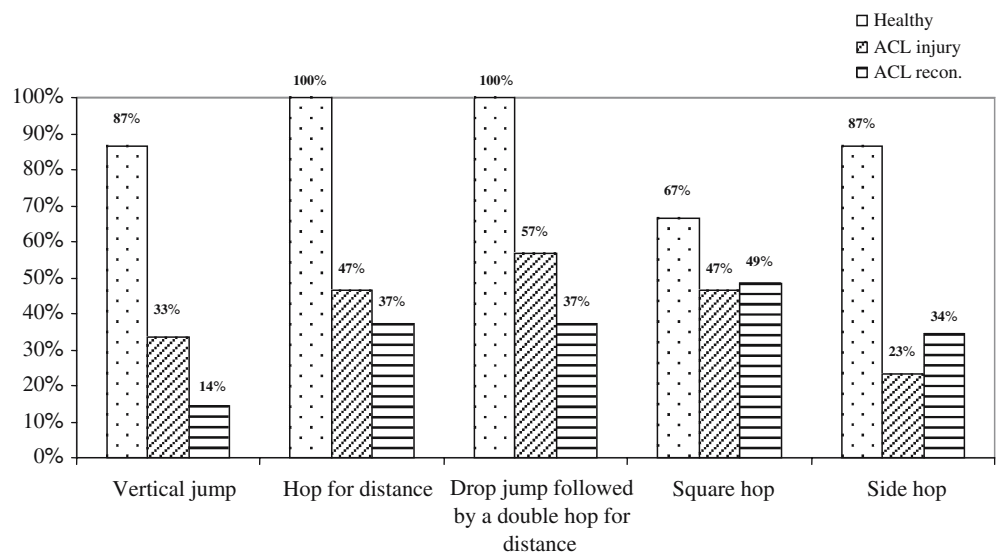
Among healthy subjects, 3 of 15 (20%) were classified as abnormal in at least one of the three tests in the test battery. Among patients with an ACL injury, 12 of 30 (40%) obtained abnormal LSI values in all three tests and 4 of 30 (13%) obtained normal values in all three tests. Twenty-six of 30 (87%) patients with an ACL injury obtained abnormal LSI values in at least one of the three tests. Among patients who had undergone ACL reconstruction, 19 of 35 (54%) obtained abnormal LSI values in all three tests and three of 35 (9%) obtained normal values in all three tests. Thirty-two of 35 (91%) patients who had undergone ACL reconstruction obtained abnormal values in at least one of the three tests.

The sensitivity of the test battery, i.e. identifying a patient as abnormal when at least one of the three tests produced an abnormal LSI value, was 87% in the patients with an ACL injury and 91% in the patients who had undergone ACL reconstruction. The accuracy of the test battery, i.e. identifying a healthy subject as normal when all three tests produced a normal LSI value and identifying a patient as abnormal when at least one of the three tests produced an abnormal LSI value, was found to be 84 and 88% respectively.

The square hop was not chosen due to its inability to discriminate between the side-to-side difference in healthy subjects and patients (Table 5). The square hop also showed the lowest specificity (67%) and accuracy (58 and 56%) of the five hop tests. The correlation with the side hop ranged from $r_s = 0.72$ to $r_s = 0.92$ ($P = 0.01$) for

Table 5 Means, standard deviations (SD) for side-to-side difference in the five hop tests

Test	Healthy subjects (<i>n</i> = 15)	ACL injury (<i>n</i> = 30)	<i>P</i> -value	ACL recon. (<i>n</i> = 35)	<i>P</i> -value
Vertical jump (cm)	1.2 ± 1.4	3.0 ± 2.5	< 0.01	4.3 ± 2.8	< 0.01
Hop for distance (cm)	6 ± 4	21 ± 24	< 0.05	20 ± 11	< 0.01
Drop jump followed by a double hop for distance (cm)	8 ± 7	45 ± 62	< 0.01	44 ± 32	< 0.01
Square hop (jumps)	4 ± 3	7 ± 12	NS	8 ± 9	NS
Side hop (jumps)	4 ± 3	11 ± 9	< 0.05	10 ± 9	< 0.05

Fig. 2 The percentage of healthy subjects (*n* = 15) and patients with ACL injury (*n* = 30) and ACL reconstruction (ACL recon.) (*n* = 35) classified as normal in the five hop tests

all subjects, indicating a strong relationship between the two endurance hop tests.

The drop jump followed by a double hop for distance was not chosen due to its high correlation with the single-leg hop for distance ($r_s > 0.80$, $P = 0.01$), accuracy similar to that of the hop for distance in the patients with an ACL injury (62 vs. 69%) and identical to that for the patients who had undergone ACL reconstruction (74%). We therefore concluded that the drop jump followed by a double hop for distance and the hop for distance were equal and measured the same ability.

Discussion

The principal finding in this study was that a test battery, consisting of the vertical jump, the hop for distance and the side hop tests, had a high ability to discriminate between the hop performance of the injured and the uninjured side both in patients 11 months after an ACL injury and in patients 6 months after ACL reconstruction. The test battery may help in the process of deciding whether and when patients can safely return to strenuous

Table 6 Results for sensitivity, specificity and accuracy in the five hop tests for healthy subjects (*n* = 15), patients with ACL injury (*n* = 30) and patients who have undergone ACL reconstruction (*n* = 35)

	Vertical jump	Hop for distance	Drop jump followed by a double hop for distance	Square hop	Side hop
Sensitivity (%)					
ACL injury	67	53	43	53	77
ACL recon.	86	63	63	51	69
Specificity (%)					
Healthy subjects	87	100	100	67	87
Accuracy (%)					
ACL injury	73	69	62	58	80
ACL recon.	86	74	74	56	74

P-values for comparisons between healthy subjects and patients with ACL injury and ACL reconstruction (ACL recon.) are given

ous physical activities after an ACL injury or reconstruction. The test battery has, however, to be studied in prospective randomised trials to be able to draw definite conclusions on its actual ability to contribute to the decision making process.

Single-leg hop tests are commonly used to study knee function in patients with an ACL injury and are designed to reflect the demands of a high level of physical activity [1–6, 8–21]. In the literature it is argued that by using a variety of hop tests, different hop qualities can be evaluated and thereby increase the opportunity to detect discrepancies in hop performance (i.e. increase the test sensitivity) [1, 4, 5, 16]. The importance of hop performance testing of patients after ACL reconstruction in a fatigued state has also been advocated [8]. In the present study three different maximum hop tests and two endurance hop tests were used.

The difference between the three maximum hop tests and the two endurance hop tests is reflected by the correlation between the two types of hop tests, indicating a fair to moderate [30] degree of relationship ($r_s=0.29$ – 0.57) in the healthy subjects. Östenberg et al. [21] reported similar correlations between the square hop and the vertical jump ($r=0.12$) and the single-leg hop for distance ($r=0.38$). The results indicate that the two types of hop tests measure different aspects of jumping ability, which is further confirmed by the factor analysis on the five hop tests. Patients with an ACL injury had, however, a strong relationship ($r_s=0.82$ – 0.88) between the two types of hop tests. One explanation could be that the patients with ACL injury in this study are not as homogeneous a group as the patients who had undergone ACL reconstruction, especially in terms of the interval between the index injury and the test occasion and in the diversity of the functional deficits.

During the two endurance hop tests, square hop and side hop, the patients performed as many jumps as possible during a period of 30 s, thereby demanding knee stability while developing muscle fatigue. The side hop test used by Itoh et al. [4] was modified not only by increasing the test duration from less than 2 to 30 s but also by increasing the distance the patients were required to jump from 30 to 40 cm. This could explain the higher sensitivity of 77% for the side-hop test found in the present study compared with the 44% in the study by Itoh et al. [4]. The reported sensitivity in these studies should, however, be compared with caution, due to different populations in terms of the time between the injury and the test occasion, for example. In this study, a sensitivity of 53% at a 90% LSI cut-off score in the single-leg hop for distance in the patients with an ACL injury was noted, which is in accordance with Noyes et al. [5] and Itoh et al. [4], who observed a sensitivity of 52 and 42%, respectively, at an 85% LSI cut-off score.

In a study by Barber et al. [1], many of the healthy subjects scored outside the normal lower limb symmetry range in the vertical jump and specificity was as low as 48% at 90% LSI). This is considerably different from the specificity of 87% in the present study, which is a more acceptable result. One explanation for this difference in results could be due to the fact that Barber et al. [1] used the jump-and-reach method, which has in another previous study showed poor reliability [coefficient of variation (CV) of 9.9%] [20].

The ICC values ranged from 0.85 to 0.97 for the five different hop tests, indicating that all the tests had high reliability. The importance of familiarisation needs to be mentioned, as a significant increase in performance was seen between test occasion 1 and 2 in the hop for distance and in the square hop. The difference may be the result of a learning process. The patients therefore practiced and were thoroughly familiarised at the physical therapy clinic before the test occasions.

The test–retest result in the vertical jump (ICC 0.97) is in agreement with those reported by Wilson et al. [31] and Manske et al. [32], who obtained ICC values of 0.97 and 0.98 using other apparatus. Several authors have described and tested the reliability of the single-leg hop for distance in healthy subjects [7, 14, 32] and patients after an ACL injury [10, 14]. ICC values ranging from 0.89 and 0.97 are reported [14, 32]. These results agree with the ICC value of 0.95 in this study.

The square hop and the side hop were generally performed only once by the healthy subjects. Despite this fact, the tests showed high reliability for both the square hop (ICC 0.85) and the side hop (ICC 0.93). Three of the patients, who had errors in more than 25% of the jumps, in one of the two tests, made a second attempt. To minimise the effect of fatigue, 3 min of rest were given before the second attempt. The number of trials was limited in order to control the fatigue effects that were induced and to reduce the total time of the test battery.

The results of the present study revealed no difference between males and females in the side-to-side comparison in any of the tests. Patients can therefore be evaluated on an individual basis, regardless of gender. This is also in accordance with previous studies [1, 4, 20].

An LSI of both 85 and 90% has been advocated to assess whether a hop test is normal or abnormal [1, 3, 8, 10, 20, 21]. Fitzgerald et al. [3] described a decision-making scheme for returning patients with an ACL injury to a high level of physical activity. Patients successfully returning to pre-injury levels of activity had a mean hop test score of 95%, compared with the mean of 85% in the patients who failed rehabilitation. On the basis of this, an LSI of 90% was chosen as the cut-off score in this study.

The square hop test used by Östenberg et al. [21] was modified in the present study. In order to force the patients to perform more demanding jumps, the volume of the square was increased. Although good reliability (ICC 0.85) was obtained in the square hop test, this test was not able to discriminate healthy subjects from patients. It was therefore concluded that the test was not optimal when it came to evaluating patients with an ACL injury or patients who had undergone ACL reconstruction. A new test, the drop jump followed by a double hop for distance, was designed for additional demands. This test is characterised by high intensity with regard to deceleration and acceleration forces. Even though the intention was to design a new, more demanding hop test, this was not supported by the results. The drop jump followed by a double hop for distance showed a strong correlation with the more established hop for distance test. The sensitivity and accuracy were not in favour of the newly designed test. To summarise, a test battery consisting of the vertical jump, the hop for distance and the side hop produced the most information. It could also be regarded as advantageous in the clinical setting to use a test battery comprising three tests, instead of four or five tests.

It is worrying to note that only one out of ten patients were classified as normal (i.e. having >90% hop performance on the injured side compared with the uninjured side) 11 months after an ACL injury and

6 months after ACL reconstruction, as it is now customary to allow a return to full sports activities 6 months after ACL reconstruction [33, 34], with some authors advocating a return to sports as early as 4 months post-operatively [35, 36].

Conclusion

The test battery consisting of both maximum single hop performances: the vertical jump and the hop for distance and hop performance while developing fatigue: the side hop, produced high test–retest reliability, sensitivity and accuracy. Further, the test battery produced higher values compared with any of the three hop tests individually, revealing that only one out of ten patients had restored hop performance 11 months after an ACL injury and 6 months after ACL reconstruction. It is concluded that this test battery showed a high ability to discriminate between the hop performance of the injured and the uninjured side both in patients with an ACL injury and in patients who have undergone ACL reconstruction.

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